

🎯 Maths Unit 3, Part 3B: Associative and distributive laws

Aasan Notes for Seemab · kitab ke page 52 se 56 tak · practice: Exercise 3.1 ke Q3 se Q7

🔗 1. Associative properties (p52, p53)

📖 Kitab ke do laws, exam mein bilkul yehi likhna hai

Associative Property of Union: $(A \cup B) \cup C = A \cup (B \cup C)$

Associative Property of Intersection: $(A \cap B) \cap C = A \cap (B \cap C)$

Teen sets par **ek hi** operation lag raha ho, to bracket chahe pehle do par lagao ya aakhri do par, jawab bilkul wohi nikalta hai. Bracket sirf yeh tay karta hai ke **kaam ka order** kya hoga, natija nahin badalta. Is liye hum **$A \cup B \cup C$** bina bracket ke bhi likh dete hain.

Proof: Associative Property of Union

Let $x \in (A \cup B) \cup C$

$\Rightarrow x \in (A \cup B)$ or $x \in C$

$\Rightarrow (x \in A \text{ or } x \in B)$ or $x \in C$

$\Rightarrow x \in A$ or $(x \in B \text{ or } x \in C)$

$\Rightarrow x \in A$ or $x \in (B \cup C)$

$\Rightarrow x \in A \cup (B \cup C)$

$\Rightarrow (A \cup B) \cup C \subseteq A \cup (B \cup C)$ (a)

Similarly, $A \cup (B \cup C) \subseteq (A \cup B) \cup C$ (b)

From (a) and (b): **$(A \cup B) \cup C = A \cup (B \cup C)$**

Proof: Associative Property of Intersection

Let $y \in (A \cap B) \cap C$

$\Rightarrow y \in A \cap B$ and $y \in C$

$\Rightarrow (y \in A \text{ and } y \in B)$ and $y \in C$

$\Rightarrow y \in A$ and $(y \in B \text{ and } y \in C)$

$\Rightarrow y \in A$ and $y \in (B \cap C)$

$\Rightarrow y \in A \cap (B \cap C)$

$\Rightarrow (A \cap B) \cap C \subseteq A \cap (B \cap C)$ (a)

Similarly, $A \cap (B \cap C) \subseteq (A \cap B) \cap C$ (b)

From (a) and (b): **$(A \cap B) \cap C = A \cap (B \cap C)$**

Poora proof likhte waqt **dono** taraf sabit karni parti hain: pehle $LHS \subseteq RHS$, phir $RHS \subseteq LHS$. Ek set doosre ke barabar tabhi hota hai jab har ek doosre ka subset ho. Sirf (a) likh dena aadha jawab hai.

🖍 Example 8 (p52, p53), Venn verification of union

$U = \{1, 2, 3, 4, 5, \dots, 15\}$, $A = \{2, 4, 6, 8, 10\}$, $B = \{1, 2, 3, 4, 5, 6, 7\}$, $C = \{2, 3, 5, 7, 11\}$.

LHS: $A \cup B = \{1, 2, 3, 4, 5, 6, 7, 8, 10\}$, so **$(A \cup B) \cup C = \{1, 2, 3, 4, 5, 6, 7, 8, 10, 11\}$**

RHS: $B \cup C = \{1, 2, 3, 4, 5, 6, 7, 11\}$, so **$A \cup (B \cup C) = \{1, 2, 3, 4, 5, 6, 7, 8, 10, 11\}$**

Dono Venn diagrams mein *same region is shaded*, is liye law verified. Bahar reh gaye 9, 12, 13, 14, 15.

🖍 Example 9 (p53), Venn verification of intersection

$U = \{1, 2, 3, \dots, 11\}$, $A = \{1, 2, 3, 5, 7, 9\}$, $B = \{1, 2, 3, 4, 5, 6\}$, $C = \{5, 6, 7, 8\}$.

LHS: $A \cap B = \{1, 2, 3, 5\}$, so **$(A \cap B) \cap C = \{5\}$** · RHS: $B \cap C = \{5, 6\}$, so **$A \cap (B \cap C) = \{5\}$**

Intersection mein hamesha **chhota** jawab aata hai; teeno circles ka beech wala hissa hi shade hoga.

🏠 2. Distributive properties (p54, p55)

📖 Kitab ke do laws

(a) Union over Intersection: $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

(b) Intersection over Union: $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

Yahan **do alag** operations hain, is liye bracket ab bekaar nahin. Bahar wala set bracket ke **dono** members par baant diya jata hai, theek jaise numbers mein $a(b + c) = ab + ac$ hota hai. Farq sirf itna ke numbers mein multiplication hi addition par distribute hota hai, ulta nahin; sets mein **dono** taraf chalta hai, union bhi aur intersection bhi. Baantne ke baad bahar wala operation andar chala jata hai aur andar wala bahar aa jata hai, yaani $\cup$ aur $\cap$ apni jagah badal lete hain.

Proof (a): Union over Intersection

Let $x \in A \cup (B \cap C)$
 $\Rightarrow x \in A$ or $x \in (B \cap C)$
 $\Rightarrow x \in A$ or $(x \in B \text{ and } x \in C)$
 $\Rightarrow (x \in A \text{ or } x \in B) \text{ and } (x \in A \text{ or } x \in C)$
 $\Rightarrow x \in (A \cup B) \text{ and } x \in (A \cup C)$
 $\Rightarrow x \in (A \cup B) \cap (A \cup C)$
 $\Rightarrow A \cup (B \cap C) \subseteq (A \cup B) \cap (A \cup C)$ (a)
 Again, let $y \in (A \cup B) \cap (A \cup C)$
 $\Rightarrow \dots \Rightarrow y \in A \cup (B \cap C)$
 $\Rightarrow (A \cup B) \cap (A \cup C) \subseteq A \cup (B \cap C)$ (b)

Proof (b): Intersection over Union

Let $x \in A \cap (B \cup C)$
 $\Rightarrow x \in A$ and $x \in (B \cup C)$
 $\Rightarrow x \in A$ and $(x \in B \text{ or } x \in C)$
 $\Rightarrow (x \in A \text{ and } x \in B) \text{ or } (x \in A \text{ and } x \in C)$
 $\Rightarrow x \in (A \cap B) \text{ or } x \in (A \cap C)$
 $\Rightarrow x \in (A \cap B) \cup (A \cap C)$
 $\Rightarrow A \cap (B \cup C) \subseteq (A \cap B) \cup (A \cap C)$ (a)
 Again, let $y \in (A \cap B) \cup (A \cap C)$
 $\Rightarrow \dots \Rightarrow y \in A \cap (B \cup C)$
 $\Rightarrow (A \cap B) \cup (A \cap C) \subseteq A \cap (B \cup C)$ (b)

Dono taraf From (a) and (b) likh kar barabari ka nishan lagta hai. Chauthi line hi asal chaal hai: or ke andar se *and* bahar nikal aata hai (aur doosre proof mein *and* ke andar se *or*).

Example 10 (p55), dono laws Venn se verify

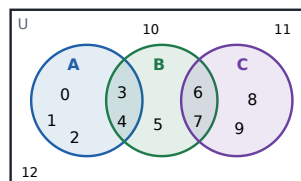
$U = \{x : x \in W \wedge x \leq 12\} = \{0, 1, 2, \dots, 12\}$, $A = \{x : x \in W \wedge x \leq 4\} = \{0, 1, 2, 3, 4\}$, $B = \{3, 4, 5, 6, 7\}$, $C = \{6, 7, 8, 9\}$.

(a) $B \cap C = \{6, 7\} \Rightarrow A \cup (B \cap C) = \{0, 1, 2, 3, 4, 6, 7\}$.

$A \cup B = \{0, 1, 2, 3, 4, 5, 6, 7\}$ and $A \cup C = \{0, 1, 2, 3, 4, 6, 7, 8, 9\} \Rightarrow (A \cup B) \cap (A \cup C) = \{0, 1, 2, 3, 4, 6, 7\}$. Verified.

(b) $B \cup C = \{3, 4, 5, 6, 7, 8, 9\} \Rightarrow A \cap (B \cup C) = \{3, 4\}$.

$A \cap B = \{3, 4\}$ and $A \cap C = \emptyset \Rightarrow (A \cap B) \cup (A \cap C) = \{3, 4\}$. Verified.



Example 10 ka naqsha: A aur B kat-te hain, B aur C kat-te hain, magar $A \cap C = \emptyset$.

⚠ Kitab ki do ghaltiyan, in par dhoka mat khana

1. Example 10 mein C ki shart ~~$\{z : z \in N \wedge 3 \leq z \leq 7\}$~~ chhapi hai, jo bilkul B jaisi hai. Naqsha saaf batata hai ke $C = \{6, 7, 8, 9\}$, yaani shart $6 \leq z \leq 9$ honi chahiye. Venn diagram ko sach maano.

2. Associative union ke proof ki step (b) mein ek bracket chhoot gaya hai: ~~$A \cup B \cup C \subseteq (A \cup B)$~~ $\cup C$. Sahi likhai hai $A \cup (B \cup C) \subseteq (A \cup B) \cup C$.

★ Key Fact box (p56), yahan se sawal bante hain

(a) $A = \{0, 1, 3, 5, 7\}$, $B = \{x \mid x \in N \wedge x \leq 5\}$ and $C = \{1, 2, 3, 4, 6, 12\}$: find $(A \cup B) \cap C$ using Venn diagrams.

(b) $A = \{0, 2, 4, 6\}$, $B = \{1, 3, 5, 7\}$ and $C = \{1, 2, 3, 6\}$: find $(A \cap B) \cap C$, $A \cap (B \cap C)$ and $(A \cup B) \cap C$ using Venn diagram.

🧠 Pakki trick

Sab se aam ghalti: $(A \cup B) \cap C$ ko $A \cup (B \cap C)$ samajh lena. Yaad rakho $(A \cup B) \cap C \neq A \cup (B \cap C)$, kyunke operations alag hain. Usool yaad rakho: **ek jaisa operation to bracket azad, alag operations to bracket pakka**. Ab Exercise 3.1 ke Q3 se Q7 khud karo.

🖋 Quiz

Ye teeno sawal tumhare Maths paper mein bhi aayenge, is liye inhein achhi tarah karo.

- Write the four laws of this section in symbols**, with their exact book names: associative property of union, associative property of intersection, distributive property of union over intersection, and distributive property of intersection over union.
- Verify, using the sets** $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5, 6\}$ and $C = \{5, 6, 7\}$: (i) $(A \cup B) \cup C = A \cup (B \cup C)$, and (ii) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$. Name the law each part verifies.
- Explain in one or two lines**. In the analytical proof we first show (a) $LHS \subseteq RHS$ and then (b) $RHS \subseteq LHS$, and only after that we write " $=$ ". Why is proving (a) alone not enough?